Superposition and interference

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add signed displacements and distinguish interference from lost energy.

SUCCESS CRITERIA - I can...

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “superposition” and “interference” to compare your model or result. Sentence starters are optional.

superposition: adding displacements or fields at the same place and time

interference: the pattern produced by overlapping waves

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

An upward rope pulse is 3 cm high before reaching an ideal free end. Is the reflected displacement upright or inverted?
Sketch it.

Word help: ideal = using stated simplifying conditions

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

Worked reasoning: At one point, two pulses have displacements +3 cm and -1 cm. Superposition gives +2 cm there. Add signed displacements, not their absolute values.

Your turn: Draw the two component displacements and the resultant at that point.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add signed displacements; label units.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two upward pulses overlap.
Sketch the combined displacement. Then sketch equal upward/downward pulses overlapping.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

Two rope pulses overlap. At point A their displacements are +3 cm and -1 cm; at point B they are +2 cm and -2 cm. Add signed displacements at each point.
Draw the two contributions and each resultant.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add signed displacements; label units.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Two rope pulses momentarily cancel. A student says the pulses are gone forever. What happens after they pass in an ideal linear medium?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

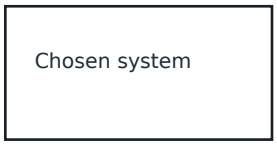
06 Review: connect an earlier idea

A system receives 60 J heat and does 40 J work on its surroundings.
Find ΔU with $\Delta U = Q - W_{by}$.
Label energy entering, leaving and stored.

Word help: surroundings = everything outside the chosen system

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

Add displacements -4 mm and +7 mm.
Name whether they reinforce or partly cancel.

Sentence starter: I notice … . This supports … because … .



Add signed displacements; label units.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Construct three different pairs of displacements with resultant zero. Can the waves still move after that instant?

Sentence starter: Both … . They differ in … because … .



Add signed displacements; label units.

Marzano check - circle: ① ② ③